

SKAGIT, WA, USA

WMO: 720272

Lat: 48.467N

Lon: 122.417W

Elev: 43

StdP: 100.81

Time Zone: -8.00 (NAP)

Period: 04-19

WBAN: 94282

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.9	-2.3	-11.0	1.5	-2.7	-7.4	2.0	-0.6	9.3	9.2	8.5	9.3	1.9	40	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	27.4	18.9	25.8	18.1	23.8	17.4	19.6	24.9	18.8	24.1	18.1	22.6	3.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	20.4	17.3	12.4	20.0	16.3	11.6	19.4	55.9	24.8	53.3	24.2	51.0	22.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	7.1	5.8	-7.5	31.5	2.1	6.0	-9.0	35.8	-10.3	39.3	-11.5	42.7	-13.0	47.1		
(5)			-8.1	21.3	2.3	1.3	-9.8	22.3	-11.1	23.1	-12.4	23.8	-14.1	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	5.1	5.7	7.6	9.5	12.7	14.8	17.2	17.5	15.2	10.6	7.3	4.7		
	DBStd	5.30	3.60	3.02	2.71	2.42	2.38	2.24	2.11	1.93	2.96	2.47	3.41	3.41		
	HDD10.0	682	154	122	83	37	4	0	0	0	0	22	92	167		
	HDD18.3	2824	409	354	332	266	175	108	48	39	101	238	331	423		
	CDD10.0	936	4	2	9	22	88	145	223	232	156	41	11	3		
	CDD18.3	37	0	0	0	0	1	3	13	13	7	0	0	0		
	CDH23.3	471	0	0	0	1	11	37	161	164	96	0	0	0		
	CDH26.7	104	0	0	0	0	1	3	29	22	48	0	0	0		
(14) Wind	WSAvg	2.5	2.8	2.7	3.1	2.7	2.3	2.2	2.2	2.1	2.0	2.3	2.7	2.6		
(15) Precipitation	PrecAvg	1105	135	87	104	90	79	64	29	36	55	112	170	142		
	PrecMax	1427	228	157	161	132	145	118	80	117	121	214	328	212		
	PrecMin	856	73	13	37	25	27	13	2	3	17	33	64	73		
	PrecStd	150	38	36	30	23	30	30	18	30	32	51	65	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	14.1	17.7	21.0	25.0	27.0	29.1	28.8	39.8	19.8	17.6	14.9		
		MCWB	11.7	9.5	11.8	14.8	17.1	18.8	19.7	19.3	11.5	14.7	13.9	12.6		
	2%	DB	12.8	12.7	15.1	17.7	22.3	24.0	27.3	27.2	25.2	17.6	14.9	12.5		
		MCWB	10.6	9.6	10.5	12.7	15.9	17.3	18.9	18.9	17.4	13.8	12.4	10.7		
	5%	DB	12.1	11.3	12.9	16.0	20.0	22.3	25.1	25.1	22.5	16.3	13.0	11.1		
		MCWB	10.4	9.0	9.8	11.4	14.9	16.7	17.7	17.8	16.6	13.2	11.2	9.5		
	10%	DB	10.2	10.0	12.3	13.8	17.8	20.3	22.9	23.0	20.8	15.0	12.3	9.1		
		MCWB	8.7	8.4	9.7	9.9	13.7	15.6	17.2	17.3	16.1	12.3	10.8	8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	11.6	13.0	15.4	17.6	20.1	21.5	20.6	19.7	16.8	14.9	13.4		
		MCDB	14.4	12.6	15.4	19.2	22.7	24.2	26.8	26.0	23.3	18.0	16.5	14.6		
	2%	WB	12.0	10.2	12.0	13.5	16.4	18.4	19.5	19.5	18.5	15.3	13.0	11.3		
		MCDB	12.7	11.4	14.0	16.9	21.1	22.8	25.1	25.3	22.8	17.0	14.6	12.2		
	5%	WB	10.3	9.2	10.9	12.1	15.3	17.2	18.7	18.8	17.3	14.0	12.0	9.4		
		MCDB	11.2	10.4	12.9	14.7	19.1	20.7	24.2	24.2	20.0	15.7	13.3	10.5		
	10%	WB	8.8	8.3	9.8	10.9	14.2	16.2	17.9	18.0	16.4	12.9	10.9	8.2		
		MCDB	9.8	9.2	12.1	13.1	17.5	19.8	22.5	22.6	19.5	14.7	12.1	9.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.1	6.9	7.8	9.2	9.3	11.1	11.2	9.4	7.7	5.9	5.1		
		MCDBR	6.4	7.8	8.9	10.4	12.9	13.0	14.5	14.5	13.1	9.2	6.9	5.8		
	5% WB	MCWBR	4.4	5.3	5.7	6.3	7.4	7.0	7.0	6.8	6.4	5.8	5.0	4.3		
		MCWBR	5.9	6.5	8.3	9.6	11.8	11.9	13.3	13.1	10.5	7.6	6.4	5.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.313	0.309	0.324	0.348	0.369	0.353	0.345	0.349	0.335	0.328	0.316	0.310		
		taud	2.536	2.562	2.538	2.460	2.416	2.477	2.526	2.521	2.555	2.568	2.546	2.500		
	Ebn at Noon	Ebn at Noon	768	861	899	904	892	908	910	892	875	824	755	717		
		Edh at Noon	60	72	86	102	111	105	99	95	84	71	59	55		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.94	1.82	2.83	4.29	5.47	5.98	6.57	5.54	3.95	2.18	1.11	0.75		
	RadStd	RadStd	0.13	0.24	0.33	0.40	0.43	0.56	0.46	0.36	0.32	0.24	0.17	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page